ANVYRA – Project Plan

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Overview

ANVYRA is a MERN web app for tracking the construction of public infrastructure in one village. It is a simplified college version of the original ANVYRA design.

Each project's Site Supervisor sends progress updates in plain words, in Hindi, Telugu or English. The app uses AI to match each update to the right task and to pick out any materials used. The Village Officer approves or rejects every update, and all progress, stock and charts are calculated from approved updates only.

The app covers four things per project: work progress, materials, issues and budget.

Roles

There are two roles. There is no worker role, since workers do the work but don't use the app.

Role	How many	Language	What they do
Village Officer	One per village	English	Sees all projects, approves or rejects updates, resolves issues, enters budget figures, adds supervisors
Site Supervisor	One per project	Hindi, Telugu or English	Sends progress updates, logs materials received, reports issues, sees the status of their own updates

A supervisor can only see and update their own project. The officer never sends updates and the supervisor never approves them.

The village and its 4 projects

The demo uses one village with four projects. Each project has its own supervisor, task list and material list. Quantities below are sample figures for the demo data.

1. Government Houses (20 houses)

Task	Unit	Planned
Foundation	houses	20
Walls up to lintel	houses	20
Roof slab	houses	20
Plastering	houses	20
Doors and windows	houses	20
Flooring	houses	20
Electrical fitting	houses	20
Handover to beneficiary	houses	20

Materials: cement (bags), steel (kg), bricks (nos), sand (cu m).

2. Overhead Water Tank

Task	Unit	Planned
Excavation	cu m	120
Foundation concrete	cu m	60
Columns	nos	8
Tank slab and walls	%	100
Pipeline laying	m	1500
Pump installation	nos	1
Leak testing	done/not done	1

Materials: cement (bags), steel (kg), sand (cu m), pipes (m).

3. Canal

Task	Unit	Planned
Earth excavation	m	2000
Canal lining	m	2000
Culverts	nos	4
Sluice gates	nos	3

Materials: cement (bags), sand (cu m), stone (cu m), steel (kg).

4. Electric Lines

Task	Unit	Planned
Pole erection	nos	80
Wire stringing	m	4000
Transformer installation	nos	2
House connections	houses	150
Meter installation	nos	150

Materials: poles (nos), wire (m), meters (nos).

Features

Progress updates

The supervisor writes what was done today in plain words, with an optional photo. The app suggests the matching task and quantity (see AI matching). The update then waits for the officer.

- **Officer's actions:** Approve, Edit (change task or quantity) or Reject with a comment.
- **Progress per task:** approved quantity ÷ planned quantity.
- **Over-planned warning:** if an update would push a task above its planned quantity, it is flagged so the officer looks carefully.

- **Supervisor's view:** each update shows as Waiting, Approved or Rejected.

Materials

Each project has a short material list with a unit for each.

- **Received:** the supervisor logs deliveries, e.g. 200 bags of cement arrived.
- **Used:** comes from the progress update text, e.g. "used 40 bags cement", and counts only once the update is approved.
- **Stock:** received minus used, calculated automatically.
- **Low stock:** a warning shows on the project card when stock falls below a set level.

Issues and delays

The supervisor reports a problem with a type (Weather, Material shortage, Labour, Land/permission, Other), a short description and an optional photo. The officer marks it Resolved with a note. Open issues show as a red count on the project card.

Budget

The officer enters each project's sanctioned amount and adds spending entries (amount, date, note). The project card shows money spent next to work done. If spending is well ahead of progress, for example 70% spent and 40% done, the card is flagged.

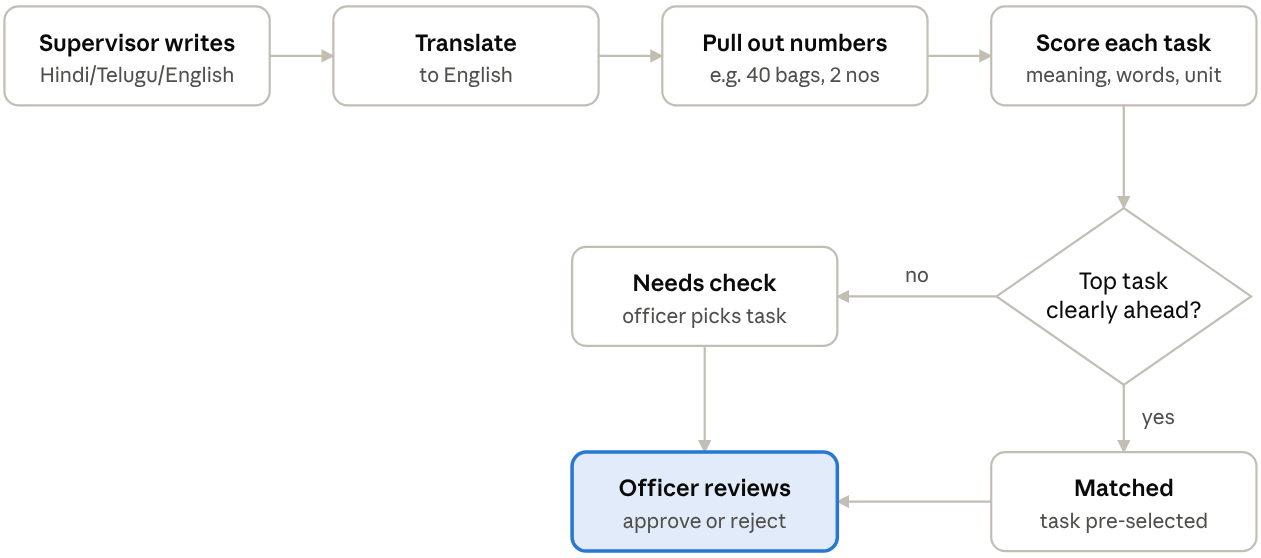
Activity log

Every important action is recorded with who did it and when: update submitted, approved, rejected, material received, issue raised or resolved. The officer can view it per project.

AI matching

Matching is the showcase feature. Because each supervisor belongs to one project, an update is only compared with that project's 4–8 tasks and 3–4 materials, so a simple scoring method is very accurate.

Every update is scored only against its own project's tasks



matching flow · 4 steps, 1 decision

The app never approves anything itself. It only pre-selects a task when it is confident, and the officer always makes the final decision.

How an update is matched

1. **Translate.** A Hindi or Telugu update is translated to English. The original text is kept and shown alongside.
2. **Split and pull out numbers.** The text is split at commas and "and", so one update can cover several tasks and materials. A number + unit is picked out of each part using a unit word list, e.g. bags/bori → bags, metre/m → m, poles/khamba → nos, houses/ghar/illu → houses.
3. **Score every task and material** on three signals, then combine them.
4. **Decide.** The update is **Matched** if the top score is at least 0.60 and at least 0.10 ahead of the second. Otherwise it is **Needs check** and the officer picks the task.
5. **Show reasons.** The officer sees the top 3 tasks with a score and why, e.g. *Columns – 91% · words: column · unit matches*.

Signal	Weight	How it works
Meaning	50%	A small sentence model (MiniLM, via `@xenova/transformers`) runs inside Node and compares meaning, so "put up poles" matches "Pole erection". Works offline, no API key
Keywords	30%	Each task has 3–6 keywords, including common Hindi and Telugu words. `fuse.js` allows for spelling mistakes
Unit	20%	Full marks if the update's unit fits the task's unit, e.g. "12 poles" fits a task measured in nos

Example

The water tank supervisor writes: *आज 2 कॉलम पूरे किए, 40 बोरी सीमेंट लगी*

- **Translated:** "Completed 2 columns today, 40 bags of cement used"
- **Part 1:** Columns, 2 nos → Matched
- **Part 2:** Cement, 40 bags → material used

Once the officer approves, Columns moves from 4/8 to 6/8 and cement stock drops by 40 bags.

Accuracy check

The demo data includes about 30 sample updates in all three languages, each with the correct task written down. A test script runs them through the matcher and reports how many were matched correctly. This gives a real accuracy number for the presentation.

Language support

Supervisor screens come in English, Hindi and Telugu; officer screens are English only.

- **Screen labels:** buttons, menus and messages on supervisor pages use `react-i18next` with three translation files. A language switch sits in the top bar and the choice is saved per user.

- **Update text:** whatever the supervisor types is translated to English by a translation service when it is submitted. The officer sees the English version with the original underneath.
- **Fallback:** if translation fails, the original text is kept and the update is marked Needs check, so nothing is lost.

Pages

The app has 11 pages: 1 shared, 6 for the officer and 4 for the supervisor.

Who	Page	What it shows
Both	Login	Email and password; sends each role to its own home page
Officer	Village Dashboard (home)	Everything at a glance, described below
Officer	Project Details	Task progress bars, materials stock, budget, issues and update history for one project
Officer	Review Updates	Pending updates with translation, photo, top 3 matches and Approve / Edit / Reject
Officer	Issues	All open and resolved issues across projects
Officer	Budget	Sanctioned vs spent per project, add spending entries
Officer	Supervisors	Add a supervisor and assign them to a project
Supervisor	My Project (home)	Their project's progress, stock and open issues
Supervisor	Send Update	Text box, optional photo, live preview of matched tasks before submitting
Supervisor	My Updates	Their updates with status and officer comments
Supervisor	Materials & Issues	Log materials received, report a new issue

Village Dashboard (officer home)

The home page should answer "how is the village doing?" in five seconds. From top to bottom:

1. **Summary strip:** overall village progress (%), updates waiting for review, open issues, total budget spent.
2. **Four project cards:** a progress ring, supervisor name, last update date, and badges for On track / Delayed, low stock and open issues. Clicking a card opens Project Details.
3. **Progress chart:** a line chart of progress over time for all 4 projects, with a dashed line for planned progress.
4. **Budget vs work chart:** a bar per project showing % spent next to % done.
5. **Updates by project:** one tab per project, each split into Waiting, Approved and Rejected.

Charts use Recharts.

Data model

Nine MongoDB collections, defined with Mongoose. Every record except User carries a `projectId`, and the API checks that a supervisor only touches their own project.

Collection	Main fields
User	name, email, passwordHash, role (officer / supervisor), language, projectId (supervisors only)
Project	name, type, village, supervisorId, startDate, endDate, sanctionedAmount, status
Task	projectId, name, unit, plannedQty, keywords[], order
Material	projectId, name, unit, keywords[], lowStockLevel
Update	projectId, supervisorId, originalText, language, englishText, photoUrl, items[] (taskId or materialId, quantity, unit, top3 matches with scores and reasons), matchStatus (Matched / Needs check), status (Waiting / Approved / Rejected), officerComment, reviewedAt

Collection	Main fields
MaterialReceipt	projectId, materialId, quantity, date, note
Issue	projectId, type, description, photoUrl, status (Open / Resolved), resolutionNote
Expense	projectId, amount, date, note

An ActivityLog collection (who, action, what, when) records the actions listed under Features.

Calculated, never stored:

- **Task progress:** sum of approved update quantities for the task ÷ plannedQty.
- **Project progress:** average of its task progress percentages.
- **Material stock:** total received − total used in approved updates.
- **Budget used:** total expenses ÷ sanctionedAmount.

These are worked out with MongoDB aggregation pipelines.

Tech stack

Layer	Choice
Frontend	React + Vite, React Router, Tailwind CSS, Recharts, react-i18next, Axios
Backend	Node.js + Express, JWT login, bcrypt for passwords, Multer for photo upload
Database	MongoDB (Atlas free tier) with Mongoose
Matching	`@xenova/transformers` (MiniLM model), `fuse.js`, a unit word list
Translation	A translation API, called only when an update is not in English
Photos	Stored in a local uploads folder, or Cloudinary when deployed

Theme

A bright, friendly countryside look across the whole site: a blue sky with soft clouds and a sun, rolling green hills, a winding path and small flowers along the bottom. The scene is drawn as an original SVG background, not a stock image, so it stays sharp on any screen.

Content sits on white, slightly transparent cards with rounded corners so text and charts stay easy to read. The colours are sky blue, grass green and warm path-yellow, with red only for warnings.

Not included

These parts of the original ANVYRA are left out to keep the project manageable:

- Primavera P6 / MS Project schedule import and schedule versions
- Worker and Project Manager roles
- OCR, PDF/Excel report reading and voice input
- Time Agent, Lessons Radar, knowledge base and AI summaries
- Delay impact preview across task dependencies
- Quality gates, inspections and reopening completed work
- Hash-chained audit log and advanced locking